A companion research article to the ProveIt Thue–Morse atlas

Thue–Morse Beyond the Atlas

Boundary Defects, Nonlinear Prouhet Geometry, and Flat Automatic q-Products

Three derived research directions, with proofs and reproducible checks

4 September 2026

Abstract

The signed Thue–Morse sequence connects binary carries, cancellation of moments, and products across geometric scales. We develop three refinements of these connections. First, a single finite boundary functional gives the exact dyadic correlation defect at every order: even correlations have an alternating defect, whereas odd correlations become an explicit telescoping difference after dyadic blocking. The same functional yields formulas for arbitrary numbers of complete blocks, without a recursive correlation hierarchy. Second, we transport the Thue–Morse signs to nonlinear Boolean-cube coordinates. An exact covering-family identity identifies the cancellation order with a hypergraph covering number for positive weights. Quadratic deformations lead to hafnians, explicit path formulas, and a monomer–dimer crossover polynomial. Third, we study the geometric product $\Phi_q(z) = \prod_{j\geq 0}(1 - zq^j)^{\varepsilon_j}$. It is flat to every algebraic order as $q \to 1^-$, but its logarithm has a nontrivial all-order saddle expansion with an explicit Lambert-W parameter and a smooth periodic amplitude. The amplitude is expressed exactly through a dyadic uniform-convolution Laplace product, making the Fabius connection precise. A joint endpoint scaling recovers classical paired products with every algebraic correction absent. The article distinguishes classical ingredients from the present derived formulas. Historical priority is not asserted: these are candidates for new results relative to the inspected atlas and the targeted literature, rather than a certified exhaustive novelty claim.

Status. Self-contained mathematical research draft prepared with ChatGPT. The main statements have proofs and independent computational checks; no claim of peer review or Lean formalization is made. The accompanying archive contains the standalone LaTeX source, PDF, commented verification program, computed tables, and source-audit notes.

Contents

1 Scope, sources, and the contribution map

1.1 What is being extended

The linked *Thue–Morse Atlas and Frontiers* in Vladimir Reshetnikov's ProveIt repository is the starting point [1]. Its product identities, finite-block calculus, moment formulas, and Fabius–Rvachev connections provide a substantial foundation. Accordingly, this article does not present the classical generating product or Prouhet cancellation as discoveries. It uses them as short, explicitly proved lemmas and develops three more specialized constructions.

The correlation literature is particularly important for avoiding overstatement. Baake and Coons show how higher-order correlations are controlled by a basic balanced two-point value, and establish the vanishing of odd-order correlations [3]. Our correlation contribution is an explicit *finite-size boundary formula*, not a claim to have discovered higher-order correlations or their basic renormalization. Likewise, the nonlinear moment section uses classical Boolean inclusion–exclusion, and the analytic section uses a classical saddle-point method. New research can lie in the exact objects these methods expose, rather than in inventing the methods themselves.

1.2 The three directions

Boundary defects. For an arbitrary finite tuple of shifts, a signed sum of at most as many terms as the largest shift determines all finite dyadic correlation corrections. For even order, the only remaining scale dependence is $(-1)^m$. For odd order, the sum over the q-th sufficiently large dyadic block is a constant multiple of $\varepsilon_{q+1} - \varepsilon_q$.

Nonlinear Prouhet geometry. The signs stay fixed, but the point with binary digits b is moved to a multilinear polynomial $x(b)$. The first surviving moment is governed by covers of the bit positions. Pair interactions produce perfect matchings and hafnians; weak pair interactions produce a full matching polynomial.

Flat automatic products. A Thue–Morse-weighted q-Pochhammer-type product tends to one faster than every power of $-\log q$. Its small deviation is nevertheless computable: a growing family of terms, centered at an explicit Lambert-W saddle, is modulated by an exact log-periodic function.

These three directions share a useful principle: the dyadic structure can often be isolated into a finite combinatorial quantity or a bounded periodic function. What remains can then be analyzed without repeatedly expanding the entire Thue–Morse word.

1.3 Novelty audit and its limitations

The source review covered the linked consolidated LaTeX article, its visible section structure and relevant passages, and targeted external searches. The reference list records primary mathematical sources. A complete recursive, commit-pinned checkout of every document in the repository was not obtained; therefore no claim of exhaustive nonduplication is justified. The consulted `main` source is mutable, and the access date is 4 September 2026.

The classical survey by Allouche and Shallit [2] supplies the background context. Generalized Prouhet constructions are discussed by Bolker, Offner, Richman, and Zara [4]. The 2026 preprint of Cloitre [9] was also screened: its transform of binary words and induced partitions provide a different direction from the fixed-sign nonlinear coordinates considered here. Products weighted by Thue–Morse signs already form an established subject [5, 6]; the distinction here is the geometric argument q^j and its singular limit. The harmonic-sum viewpoint of Flajolet, Gourdon, and Dumas [7] provides analytic context.

The results below are thus presented as *proved derived refinements and novelty candidates*, with the exact formulas available for further literature comparison. They should not be advertised as the first results of their kind without an additional specialist priority review.

2 Binary and analytic preliminaries

We use zero-based indexing. For $n \geq 0$, let $s_2(n)$ be the number of ones in its binary expansion and define

$$\varepsilon_n = (-1)^{s_2(n)}. \tag{2.1}$$

Thus the first sixteen signs are

$$1, -1, -1, 1, -1, 1, 1, -1, -1, 1, 1, -1, 1, -1, -1, 1.$$

Empty products are one. A repeated shift in a correlation is permitted; its multiplicity is part of the definition.

Lemma 2.1 (Block and reflection identities)**.** *For $N = 2^m$, $q \geq 0$, and $0 \leq r < N$,*

$$\varepsilon_{qN+r} = \varepsilon_q \varepsilon_r, \qquad \varepsilon_{N-1-r} = (-1)^m \varepsilon_r. \tag{2.2}$$

If $m \geq 1$, then, with the index interpreted modulo N,

$$\varepsilon_{(r+N/2) \bmod N} = -\varepsilon_r. \tag{2.3}$$

Proof. The binary digits of qN and r occupy disjoint positions, so their digit sums add. Subtraction from $N-1$ complements exactly m digits, proving the second formula. Adding $N/2$ modulo N toggles the highest of those digits and leaves the others unchanged, proving the third. □

Lemma 2.2 (The generating product)**.** *For $|z| < 1$,*

$$E(z) := \sum_{n \geq 0} \varepsilon_n z^n = \prod_{j \geq 0} (1 - z^{2^j}), \qquad E(z) = (1-z)E(z^2). \tag{2.4}$$

The product converges locally uniformly in the unit disk.

Proof. In the finite product through $j = m-1$, choosing the factor $-z^{2^j}$ exactly for the one-bits of $n < 2^m$ gives the coefficient ε_n. Hence $\prod_{j<m}(1 - z^{2^j}) = \sum_{n<2^m} \varepsilon_n z^n$. For $|z| \leq \rho < 1$, $\sum_j \rho^{2^j} < \infty$, so the product converges uniformly there. Passing to the limit gives the identity. Removing its first factor gives the functional equation. □

2.1 An exact adjacent-correlation sum

The following elementary identity will replace an infinite correlation recursion by a finite binary-digit expression.

Definition 2.3. For an integer $Q = \sum_{j \geq 0} b_j 2^j \geq 0$, define the alternating digit sum

$$\sigma(Q) = \sum_{j \geq 0} (-1)^j b_j. \tag{2.5}$$

In particular, $\sigma(0) = 0$ and $\sigma(2^k) = (-1)^k$.

Lemma 2.4 (Adjacent correlation at every endpoint)**.** *For every* $Q \ge 0$*,*

$$C(Q) := \sum_{q=0}^{Q-1} \varepsilon_q \varepsilon_{q+1} = -\frac{Q}{3} - \frac{2}{3}\sigma(Q). \tag{2.6}$$

Consequently $C(Q)/Q \to -1/3$ *as* $Q \to \infty$*.*

Proof. Write $c(q) = \varepsilon_q \varepsilon_{q+1}$. The binary recursion gives $c(2q) = -1$ and $c(2q+1) = -c(q)$. Therefore

$$C(2Q) = -Q - C(Q), \qquad C(2Q+1) = -Q - C(Q) - 1.$$

On the other hand, $\sigma(2Q) = -\sigma(Q)$ and $\sigma(2Q+1) = 1 - \sigma(Q)$. Substitution shows that the proposed right-hand side satisfies the same recurrences and agrees at zero; induction proves the formula. The number of nonzero summands in $\sigma(Q)$ is at most the binary length of Q, which is $O(\log(Q+1))$. Division by Q proves the limit. □

3 Exact boundary defects for correlations of every order

3.1 The finite boundary functional

Fix a tuple

$$\boldsymbol{h} = (h_1, \ldots, h_p), \qquad p \ge 1, \qquad 0 = \min_i h_i, \qquad H = \max_i h_i.$$

For $N = 2^m > H$, define the noncyclic and cyclic sums

$$S_m(\boldsymbol{h}) = \sum_{n=0}^{N-1} \prod_{i=1}^{p} \varepsilon_{n+h_i}, \tag{3.1}$$

$$R_m(\boldsymbol{h}) = \sum_{n=0}^{N-1} \prod_{i=1}^{p} \varepsilon_{(n+h_i) \bmod N}. \tag{3.2}$$

The noncyclic sum uses the infinite sequence even when a shifted index exits the first block. The distinction between these two conventions is essential.

For each $1 \le t \le H$, set

$$b_t = \#\{i : h_i \ge t\}, \tag{3.3}$$

$$G_t(\boldsymbol{h}) = \prod_{h_i < t} \varepsilon_{t-h_i-1} \prod_{h_i \ge t} \varepsilon_{h_i - t}. \tag{3.4}$$

Define the *boundary functional*

$$D(\boldsymbol{h}) = \sum_{\substack{1 \le t \le H \\ b_t \text{ odd}}} G_t(\boldsymbol{h}). \tag{3.5}$$

It is an integer and satisfies $|D(\boldsymbol{h})| \le H$. All its sign indices lie between zero and $H-1$, so it is independent of the dyadic level m.

Theorem 3.1 (One boundary functional, all dyadic levels)**.** *Suppose* $m \ge 1$ *and* $N = 2^m > H$*.*

If p *is even, the correlation mean*

$$\eta(\boldsymbol{h}) = \lim_{X \to \infty} \frac{1}{X} \sum_{n=0}^{X-1} \prod_i \varepsilon_{n+h_i} \tag{3.6}$$

exists, and

$$S_m(\boldsymbol{h}) = N\eta(\boldsymbol{h}) - \frac{2}{3}(-1)^m D(\boldsymbol{h}), \tag{3.7}$$

$$R_m(\boldsymbol{h}) = N\eta(\boldsymbol{h}) + \frac{4}{3}(-1)^m D(\boldsymbol{h}). \tag{3.8}$$

If p is odd, the same mean exists and equals zero, while

$$S_m(\boldsymbol{h}) = -2D(\boldsymbol{h}), \qquad R_m(\boldsymbol{h}) = 0. \tag{3.9}$$

Proof. Write an index as $n = qN + r$, with $0 \le r < N$, and let

$$\chi_i(r) = \mathbf{1}_{\{r+h_i \ge N\}}, \quad b(r) = \sum_i \chi_i(r), \quad g(r) = \prod_i \varepsilon_{(r+h_i) \bmod N}.$$

Because $h_i < N$, each shifted index crosses at most one block boundary. The block identity (2.2) gives the exact factorization

$$\prod_i \varepsilon_{qN+r+h_i} = g(r)\varepsilon_q^{p-b(r)}\varepsilon_{q+1}^{b(r)}. \tag{3.10}$$

Separate the local sums according to the parity of the crossing number:

$$U_m = \sum_{b(r)\text{ even}} g(r), \qquad V_m = \sum_{b(r)\text{ odd}} g(r).$$

Thus $R_m = U_m + V_m$.

We first compute V_m. An odd crossing number can occur only at $r = N - t$, $1 \le t \le H$. A noncrossing index then has the form $N - (t - h_i)$; reflection in (2.2) turns its sign into $(-1)^m \varepsilon_{t-h_i-1}$. A crossing index has the local value $h_i - t$. There are $p - b_t$ noncrossing indices. On the terms selected for V_m, b_t is odd, so

$$V_m = \begin{cases} (-1)^m D(\boldsymbol{h}), & p \text{ even}, \\ D(\boldsymbol{h}), & p \text{ odd}. \end{cases} \tag{3.11}$$

Suppose now that p is even. Summing (3.10) over one block produces

$$\sum_{r=0}^{N-1} \prod_i \varepsilon_{qN+r+h_i} = U_m + V_m \varepsilon_q \varepsilon_{q+1}. \tag{3.12}$$

Over Q complete blocks, division by QN and use of (2.6) give $\eta = (U_m - V_m/3)/N$. An arbitrary endpoint leaves fewer than N terms, whose contribution to the average tends to zero. This proves existence of the full limit. At $q = 0$, $\varepsilon_0 \varepsilon_1 = -1$, hence $S_m = U_m - V_m$. Combining this with $R_m = U_m + V_m$, the formula for η, and (3.11) proves (3.7)–(3.8).

If p is odd, the cyclic sum is zero: shifting every summation index by $N/2$ reverses all p signs by (2.3). Consequently $U_m = -V_m = -D(\boldsymbol{h})$. Formula (3.10) now gives

$$\sum_{r=0}^{N-1} \prod_i \varepsilon_{qN+r+h_i} = D(\boldsymbol{h})(\varepsilon_{q+1} - \varepsilon_q). \tag{3.13}$$

This telescopes over q, so the sum over any number of complete blocks is bounded by $2|D|$. A partial final block has fewer than N terms. The correlation mean is therefore zero. Taking $q = 0$ proves (3.9). □

Remark 3.2 (What is and is not new in this statement). The means and their renormalization belong to the established correlation theory [3]. The refinement emphasized here is the explicit finite functional (3.5), its independence of m, and the exact boundary and telescoping formulas. The proof also shows why the number of correlation factors does not introduce additional dyadic defect modes: once a block is longer than every shift, the only coarse signs available are ε_q and ε_{q+1}.

3.2 All block multiples and aligned intervals

Write

$$S_{m,Q}(\boldsymbol{h}) = \sum_{n=0}^{QN-1} \prod_i \varepsilon_{n+h_i}, \qquad Q \geq 0.$$

Corollary 3.3 (Exact endpoint formulas)**.** *Under the assumptions of Theorem 3.1,*

$$S_{m,Q}(\boldsymbol{h}) = \begin{cases} QN\eta(\boldsymbol{h}) - \dfrac{2}{3}(-1)^m D(\boldsymbol{h})\sigma(Q), & p \text{ even}, \\ D(\boldsymbol{h})(\varepsilon_Q - 1), & p \text{ odd}. \end{cases} \tag{3.14}$$

In particular, the sum over the aligned interval $[AN, BN)$ is obtained by subtracting the same expression at $Q = A$ from that at $Q = B$.

Proof. For even p, sum (3.12) and use $U_m = N\eta + V_m/3$, (2.6), and (3.11). For odd p, telescope (3.13). Interval subtraction is exact. □

Thus finite endpoints are controlled either by the alternating digit sum of Q or by its Thue–Morse sign. For a fixed tuple, the odd-order partial sums are bounded uniformly in the endpoint. Even-order centered partial sums have an elementary logarithmic bound.

Corollary 3.4 (Discrepancy bounds)**.** *For fixed $\boldsymbol{h}$, let $X = QN + R$, $0 \leq R < N$. If p is even, then*

$$\left| \sum_{n<X} \prod_i \varepsilon_{n+h_i} - X\eta(\boldsymbol{h}) \right| \leq \frac{2}{3}|D(\boldsymbol{h})|\,|\sigma(Q)| + 2N. \tag{3.15}$$

If p is odd, then

$$\left| \sum_{n<X} \prod_i \varepsilon_{n+h_i} \right| \leq 2|D(\boldsymbol{h})| + N. \tag{3.16}$$

Proof. Apply (3.14). A partial block contains at most N signs of absolute value one, and $|\eta| \leq 1$, so centering that partial block costs at most $2N$. In the odd case no centering is necessary. □

The even bound is $O_{\boldsymbol{h}}(\log(X+2))$, while the odd bound is $O_{\boldsymbol{h}}(1)$. Neither statement is asserted to have an optimal constant. For shifts growing with X, the explicit dependence on H and N must be retained; a fixed-shift estimate must not be used as a uniform growing-shift estimate.

3.3 Two-level extraction and rationality

Corollary 3.5 (Exact extrapolation, not an asymptotic approximation)**.** *If p is even and $N = 2^m > H$,*

$$\eta(\boldsymbol{h}) = \frac{S_m(\boldsymbol{h}) + S_{m+1}(\boldsymbol{h})}{3N} = \frac{R_m(\boldsymbol{h}) + R_{m+1}(\boldsymbol{h})}{3N}. \tag{3.17}$$

Moreover,

$$\eta(\boldsymbol{h}) \in \frac{1}{3N}\mathbb{Z}, \qquad \left| \frac{S_m}{N} - \eta \right| \leq \frac{2H}{3N}, \qquad \left| \frac{R_m}{N} - \eta \right| \leq \frac{4H}{3N}. \tag{3.18}$$

Proof. The defects in consecutive dyadic levels have opposite signs and equal magnitudes. Adding the two exact formulas cancels them, while the main terms sum to $3N\eta$. The rationality and inequalities follow from integrality of S_m, D and $|D| \leq H$. □

The frequency module with denominators consisting of a factor three and powers of two is already part of the classical structure [3]. Here it is recovered together with an explicit finite extraction formula.

3.4 Pairs, a convolution coefficient, and examples

For $\boldsymbol{h} = (0, h)$, the odd crossing number is one throughout the boundary, so

$$D(0,h) = \sum_{j=0}^{h-1} \varepsilon_j \varepsilon_{h-1-j} = [z^{h-1}]E(z)^2 \qquad (h \geq 1). \tag{3.19}$$

This is a useful bridge between a finite correlation defect and the ordinary generating series.

Proposition 3.6 (Binary recursion for the pair defect)**.** *With $D_h = D(0,h)$, set $D_0 = 0$. Then $D_1 = 1$, and*

$$D_{2h} = -2D_h, \qquad D_{2h+1} = D_h + D_{h+1}. \tag{3.20}$$

Proof. Write $a_n = [z^n]E(z)^2$, with $a_{-1} = 0$. Squaring $E(z) = (1-z)E(z^2)$ gives $a_{2n} = a_n + a_{n-1}$ and $a_{2n+1} = -2a_n$. Since $D_h = a_{h-1}$, the claimed formulas follow, including the boundary values. □

Table 1: Exact even-order examples. Here $N = 2^m > H$, and cyclic and noncyclic sums use the conventions in (3.1)–(3.2).

Shifts $\boldsymbol{h}$	m	D	η	S_m	R_m
$(0,1)$	1	1	$-1/3$	0	-2
$(0,2)$	2	-2	$-1/3$	0	-4
$(0,3)$	2	-1	$1/3$	2	0
$(0,1,2,3)$	2	2	$1/3$	0	4
$(0,1,3,7)$	3	-3	0	-2	4
$(0,2,5,9)$	4	-2	$1/6$	4	0
$(0,1,2,4,7,11)$	4	-5	$-1/3$	-2	-12

Example 3.7 (A zero mean with a persistent dyadic boundary signal)**.** For $\boldsymbol{h} = (0, 1, 3, 7)$, Table 1 gives $\eta = 0$, $D = -3$. For every $m \geq 3$,

$$S_m = 2(-1)^m, \qquad R_m = -4(-1)^m.$$

The unnormalized finite signal does not tend to zero even though its normalized mean does. Cyclic and noncyclic experiments detect different boundary signals.

Example 3.8 (An odd-order telescoping observable)**.** For $\boldsymbol{h} = (0, 1, 2)$, $D = -1$. Every dyadic block of length $N \geq 4$ satisfies

$$\sum_{r=0}^{N-1} \varepsilon_{qN+r}\varepsilon_{qN+r+1}\varepsilon_{qN+r+2} = \varepsilon_q - \varepsilon_{q+1}.$$

Thus the sum up to QN is exactly $1 - \varepsilon_Q$, taking only the values zero and two. This is stronger information than vanishing of the limiting mean.

3.5 A nonrecursive algorithm

To evaluate an even correlation mean, choose $m = \max(1, \lceil \log_2(H+1) \rceil)$. Compute one block sum and the boundary functional, then use

$$\eta(\boldsymbol{h}) = \frac{3S_m(\boldsymbol{h}) + 2(-1)^m D(\boldsymbol{h})}{3 \cdot 2^m}.$$

For fixed order p, a direct implementation uses $O(pH)$ sign evaluations, up to the harmless factor coming from $2^m < 2(H+1)$. The integer digit-parity operation makes each sign inexpensive. This is a transparent alternative to recursively reducing a large family of correlation tuples. It is not a claim that this direct algorithm beats every specialized recursion.

4 Nonlinear Prouhet cancellation as a covering problem

4.1 Moving the points while keeping the signs

The usual Prouhet sum assigns the sign $(-1)^{b_0+\cdots+b_{m-1}}$ to the point $\sum_i 2^i b_i$. We retain the signs but allow nonlinear coordinates. Let $V = \{0, \ldots, m-1\}$, let $\mathcal{H}$ be a finite family of nonempty subsets of V, and put

$$x(b) = c_0 + \sum_{S\in\mathcal{H}} c_S b_S, \qquad b_S = \prod_{i\in S} b_i, \qquad b \in \{0,1\}^m. \tag{4.1}$$

Terms with zero coefficient can be removed. Every function on the Boolean cube has a unique multilinear representation, but sparsity of $\mathcal{H}$ is what makes the following description informative.

Define the signed exponential generating function

$$Z_x(t) = \sum_{b\in\{0,1\}^m} (-1)^{|b|} \mathrm{e}^{tx(b)} = \sum_{r\ge 0} \frac{t^r}{r!} \sum_{b\in\{0,1\}^m} (-1)^{|b|} x(b)^r. \tag{4.2}$$

A subfamily $\mathcal{C} \subseteq \mathcal{H}$ is a *cover* if $\bigcup_{S\in\mathcal{C}} S = V$. Let $\tau(\mathcal{H})$ be the minimum number of members in a cover, and set $\tau = \infty$ if there is no cover.

Theorem 4.1 (Exact covering-family identity)**.** *For every complex t and all complex coefficients in (4.1),*

$$Z_x(t) = (-1)^m \mathrm{e}^{tc_0} \sum_{\substack{\mathcal{C}\subseteq\mathcal{H} \\ \bigcup_{S\in\mathcal{C}} S = V}} \prod_{S\in\mathcal{C}} (\mathrm{e}^{tc_S} - 1). \tag{4.3}$$

Proof. On the Boolean cube, b_S takes only the values zero and one, hence

$$\mathrm{e}^{tc_S b_S} = 1 + (\mathrm{e}^{tc_S} - 1) b_S.$$

Multiply these identities over $S \in \mathcal{H}$. Choosing the nonconstant term for the subfamily $\mathcal{C}$ contributes $\prod_{S\in\mathcal{C}} (\mathrm{e}^{tc_S} - 1) b_U$, where $U = \bigcup_{S\in\mathcal{C}} S$; repeated bit factors collapse because $b_i^2 = b_i$. The signed cube sum of b_U factors over coordinates. If any coordinate is missing from U, its factor is $1 - 1 = 0$. If $U = V$, only the all-one vertex contributes, and its sign is $(-1)^m$. Multiplying by e^{tc_0} proves (4.3). ☐

Corollary 4.2 (Cover number and the first surviving moment)**.** *If $\tau < \infty$, then*

$$\sum_b (-1)^{|b|} x(b)^r = 0 \quad (0 \le r < \tau), \tag{4.4}$$

and

$$\sum_b (-1)^{|b|} x(b)^\tau = (-1)^m \tau! \sum_{\substack{\mathcal{C}\ \text{cover} \\ |\mathcal{C}|=\tau}} \prod_{S\in\mathcal{C}} c_S. \tag{4.5}$$

If all $c_S > 0$, this is nonzero, so the vanishing order of Z_x at zero is exactly τ. If there is no cover, then $Z_x \equiv 0$.

Proof. Every factor $\mathrm{e}^{tc_S} - 1$ begins with tc_S, so a cover with k members contributes first at order t^k. At the minimum order only its linear terms contribute. The constant-coordinate factor starts with one and does not affect that coefficient. For positive weights all minimum-cover contributions have the same sign. If there is no cover, the sum in (4.3) is empty. ☐

Remark 4.3 (A cancellation caveat). For negative or complex weights, the minimum-cover sum may vanish. Then τ is a lower bound on the vanishing order, not necessarily its exact value. This distinction matters when designing enhanced cancellation by tuning interaction coefficients.

4.2 All moments in a finite closed form

Extracting coefficients from (4.3) gives a formula that involves no auxiliary moment recurrence:

$$\sum_b (-1)^{|b|} x(b)^r = (-1)^m r! \sum_{\mathcal{C}\text{ cover}} \sum_{\substack{k_0 \ge 0,\ k_S \ge 1\ (S\in\mathcal{C}) \\ k_0 + \sum_{S\in\mathcal{C}} k_S = r}} \frac{c_0^{k_0}}{k_0!} \prod_{S\in\mathcal{C}} \frac{c_S^{k_S}}{k_S!}. \tag{4.6}$$

The inner sum is finite for each r; covers with more than r members can be discarded. The proof is simply the absolutely convergent exponential series expansion of the finite expression (4.3).

Corollary 4.4 (A degree bound, with a sharper sparse refinement)**.** *If every edge $S \in \mathcal{H}$ has at most d vertices, then*

$$\sum_b (-1)^{|b|} x(b)^r = 0 \quad \textit{whenever} \quad r < \left\lceil \frac{m}{d} \right\rceil. \tag{4.7}$$

The exact cover number can be larger than this degree bound.

Proof. A union of r edges of size at most d contains at most rd vertices. It cannot cover V when $rd < m$. Apply Corollary 4.2. □

For the ordinary linear coordinate $x(b) = c_0 + \sum_i a_i b_i$, the only cover uses all singleton edges. Formula (4.3) becomes

$$Z_x(t) = (-1)^m \mathrm{e}^{tc_0} \prod_{i=0}^{m-1} (\mathrm{e}^{ta_i} - 1), \tag{4.8}$$

recovering classical Prouhet cancellation and the leading moment $(-1)^m m! \prod_i a_i$. This recovery is a consistency check, not a new version of the classical theorem.

4.3 Quadratic geometry and hafnians

Consider

$$x(b) = c_0 + \sum_{i\in V} a_i b_i + \sum_{i<j} J_{ij} b_i b_j, \tag{4.9}$$

where the symmetric matrix J has zero diagonal. For an even set of vertices U, define its hafnian by

$$\mathrm{Haf}(J_U) = \sum_{M \text{ perfect matching of } U} \prod_{\{i,j\}\in M} J_{ij}, \qquad \mathrm{Haf}(J_\varnothing) = 1. \tag{4.10}$$

A perfect matching partitions the vertices into disjoint pairs. Zero entries of J simply remove the corresponding edges.

Theorem 4.5 (Quadratic leading moments)**.** *For $m = 2r$, all moments of degree less than r vanish, and*

$$\sum_b (-1)^{|b|} x(b)^r = r! \,\mathrm{Haf}(J). \tag{4.11}$$

For $m = 2r+1$, all moments of degree less than $r+1$ vanish, and

$$\begin{aligned} \sum_b (-1)^{|b|} x(b)^{r+1} = -(r+1)! \Bigg[&\sum_i a_i \,\mathrm{Haf}(J_{V\setminus\{i\}}) \\ &+ \sum_{i<j<k} (J_{ij}J_{ik} + J_{ij}J_{jk} + J_{ik}J_{jk}) \,\mathrm{Haf}(J_{V\setminus\{i,j,k\}}) \Bigg]. \end{aligned} \tag{4.12}$$

The second sum is empty when $m = 1$.

Proof. For $m = 2r$, a cover using r sets of size at most two must use only pairs, with no repeated vertices. These covers are precisely perfect matchings. The factor $(-1)^m$ is positive. This proves (4.11).

For $m = 2r + 1$, a cover with $r + 1$ members has only two possible shapes. It can use one singleton and r disjoint pairs, or it can use $r + 1$ pairs with exactly one repeated vertex. In the latter case, two pairs form a path on three vertices, while all remaining pairs form a perfect matching on the complement. A three-element vertex set has three possible two-edge paths, with weights displayed in (4.12). There can be no other shape: the maximum total incidence count is $2r+2$, so covering $2r+1$ vertices permits only one incidence in excess of a disjoint cover. The sign $(-1)^{2r+1}$ is negative. The degree bound and Corollary 4.2 complete the proof. □

For even m, the first potentially nonzero moment is completely independent of the linear coefficients and the translation c_0. The quadratic interaction graph alone controls it. This makes a hafnian an observable of a nonlinearly deformed Thue–Morse partition.

4.4 An explicit nearest-neighbor family

Set

$$x_\lambda(b) = \sum_{i=0}^{m-1} 2^i b_i + \lambda \sum_{i=0}^{m-2} b_i b_{i+1}, \qquad \lambda > 0. \tag{4.13}$$

The interaction graph is a path, and every allowed coefficient is positive.

Corollary 4.6 (Exact path moments)**.** *If $m = 2r$, the first nonzero moment has degree r, and*

$$\sum_b (-1)^{|b|} x_\lambda(b)^r = r!\lambda^r. \tag{4.14}$$

If $m = 2r + 1$, the first nonzero moment has degree $r + 1$, and

$$\sum_b (-1)^{|b|} x_\lambda(b)^{r+1} = -(r+1)! \left[\lambda^r \frac{4^{r+1}-1}{3} + r\lambda^{r+1}\right]. \tag{4.15}$$

Proof. An even path has a unique perfect matching, proving (4.14). For an odd path, removal of a singleton leaves two even paths precisely when the removed vertex has even index $2j$, $0 \le j \le r$. The corresponding linear coefficients sum to $\sum_{j=0}^{r} 2^{2j} = (4^{r+1} - 1)/3$, and the matching weight is λ^r.

A two-edge path that leaves an even path on both sides must start at an even index $0, 2, \ldots, 2r - 2$. There are r choices. Including the matching on the complement, each such cover has weight λ^{r+1}. Theorem 4.5 gives the formula. Positivity ensures that these leading moments do not vanish. □

At $\lambda = 0$, the coordinate is linear and has cancellation through degree $m - 1$. For every $\lambda > 0$, the first surviving degree falls to $\lceil m/2 \rceil$. Thus the *order* of cancellation is not stable under arbitrarily small quadratic perturbations, even though the newly surviving moments can have small coefficients.

For example, when $m = 4$,

$$\sum_{n=0}^{15} \varepsilon_n x_\lambda(n)^2 = 2\lambda^2,$$

where $x_\lambda(n)$ means (4.13) evaluated at the four binary digits of n. This is an exact polynomial identity, not merely a small-λ approximation.

5 Weak interactions and a matching-polynomial crossover

The previous phenomenon has a useful scaling resolution. Let the quadratic coupling shrink at the same rate as the generating-function variable. Define

$$x_{\theta t}(b) = c_0 + \sum_i a_i b_i + \theta t \sum_{i<j} J_{ij} b_i b_j. \tag{5.1}$$

A singleton now contributes one power of t to the exponent, while an edge contributes two. This balances its ability to cover two bit positions.

Definition 5.1 (Weighted matching polynomial)**.** Let G be the graph of nonzero entries of J. Set

$$\mathcal{M}_G(\theta; \boldsymbol{a}, J) = \sum_{M \text{ matching in } G} \theta^{|M|} \prod_{\{i,j\}\in M} J_{ij} \prod_{i \text{ unmatched by } M} a_i. \tag{5.2}$$

Here a matching need not cover all vertices; unmatched vertices carry their singleton weights.

Theorem 5.2 (The weak-coupling crossover)**.** *For fixed coefficients and m, as $t \to 0$,*

$$\sum_{b\in\{0,1\}^m} (-1)^{|b|} \exp(t x_{\theta t}(b)) = (-1)^m t^m \mathcal{M}_G(\theta; \boldsymbol{a}, J) + O(t^{m+1}), \tag{5.3}$$

uniformly for θ in a compact subset of $\mathbb{C}$.

Proof. Apply the covering identity to singleton factors $e^{ta_i} - 1 = ta_i + O(t^2)$ and pair factors $e^{\theta t^2 J_{ij}} - 1 = \theta t^2 J_{ij} + O(t^4)$. If a cover has s singleton members and k pair members, its first possible order is $s + 2k$. A cover of all m vertices has $s + 2k \geq m$, with equality exactly when all its members are disjoint. Such covers consist of a matching and singleton choices for its unmatched vertices. Their leading coefficients give (5.2). All other terms have order at least $m + 1$. There are finitely many covers, and their Taylor estimates are uniform for bounded θ, giving the stated remainder. □

This polynomial simultaneously records the purely linear Prouhet coefficient $\prod_i a_i$ at $\theta = 0$ and, for even m, the hafnian coefficient of $\theta^{m/2}$. It provides a precise transition between the two cancellation regimes.

5.1 A path recurrence and a complete-graph formula

For a path on vertices $0, \dots, k-1$, let M_k denote (5.2). Then

$$M_0 = 1, \quad M_1 = a_0, \quad M_k = a_{k-1} M_{k-1} + \theta J_{k-2,k-1} M_{k-2} \quad (k \geq 2). \tag{5.4}$$

Indeed, the last vertex is either unmatched or paired with its only available neighbor. These alternatives are disjoint and exhaustive.

If the graph is complete and all $a_i = a$, $J_{ij} = J$, then

$$M_m = \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{m!}{2^k k!(m-2k)!} (\theta J)^k a^{m-2k}. \tag{5.5}$$

Choose $2k$ matched vertices and partition them into k unordered pairs; the number of choices is the coefficient displayed. Equivalently,

$$\sum_{m\geq 0} M_m \frac{u^m}{m!} = \exp\left(au + \frac{\theta J u^2}{2}\right). \tag{5.6}$$

No square-root convention or special-function normalization is needed for this finite formula.

5.2 Cancellation design and its limits

The positivity assumption in Corollary 4.2 is useful for proving exact orders, but it rules out cancellations among minimum covers. Allowing signed interactions changes the problem: one can seek a nonzero matrix J with $\mathrm{Haf}(J) = 0$, thereby suppressing the first quadratic moment. For four vertices the condition is simply

$$J_{01}J_{23} + J_{02}J_{13} + J_{03}J_{12} = 0.$$

This is a concrete algebraic design equation. Satisfying it suppresses the moment of degree two, but does not by itself suppress the degree-three moment; that requires the next coefficient of (4.3). The covering identity therefore provides a hierarchy of *finite polynomial constraints*, rather than an unjustified promise that one canceled hafnian restores full Prouhet regularity.

6 A Thue–Morse-weighted geometric product

6.1 Definition and exact logarithmic series

For $0 < q < 1$ and $|z| < 1$, define

$$\Phi_q(z) = \prod_{j=0}^{\infty}(1 - zq^j)^{\varepsilon_j}. \tag{6.1}$$

This resembles the ordinary q-Pochhammer product, but each factor is put in the numerator or denominator according to a Thue–Morse sign. It is not a fractional-power product: the exponents are the integers ± 1. On the unit disk, we use the logarithm specified by the Taylor series at $z = 0$.

Theorem 6.1 (Convergence, a dyadic equation, and a positive series)**.** *The product* (6.1) *converges locally uniformly in* $|z| < 1$*, defines a nonvanishing holomorphic function there, and satisfies*

$$\log \Phi_q(z) = -\sum_{r\geq 1} \frac{z^r}{r} E(q^r), \tag{6.2}$$

$$\Phi_q(z) = \frac{\Phi_{q^2}(z)}{\Phi_{q^2}(qz)}. \tag{6.3}$$

For $0 < z < 1$*, one has* $0 < \Phi_q(z) < 1$*.*

Proof. For $|z| \leq \rho < 1$,

$$\sum_{j\geq 0} |\log(1 - zq^j)| \leq \frac{\rho}{1-\rho} \sum_{j\geq 0} q^j < \infty.$$

The logarithmic sum therefore converges uniformly, and its exponential is a nonzero holomorphic function. Expanding each logarithm is justified by

$$\sum_{j\geq 0}\sum_{r\geq 1} \frac{|z|^r q^{jr}}{r} = \sum_{r\geq 1} \frac{|z|^r}{r(1-q^r)} < \infty.$$

Interchanging sums and using (2.4) gives (6.2). Separating even and odd values of j in the original product gives (6.3). Finally, every factor of $E(q^r) = \prod_{k\geq 0}(1 - q^{r2^k})$ lies in $(0, 1)$, and its convergent product is positive. Thus the series on the right of (6.2) is strictly negative for real $0 < z < 1$, proving the inequalities. □

The sign-weighted product literature includes extensive families in which the factor is a rational function of the index [5, 6]. Formula (6.1) instead samples a geometric progression. Our purpose is not to reclassify all such products, but to resolve its singular limit at $q = 1$.

6.2 Two useful truncation bounds

The defining logarithmic product, stopped before $j = J$, has an error bounded by

$$\left|\sum_{j\ge J} \varepsilon_j \log(1 - zq^j)\right| \le \frac{|z|q^J}{(1-|z|)(1-q)}. \tag{6.4}$$

This follows from the same absolute logarithm bound as in the proof above. For the series (6.2), real $q \in (0,1)$ gives

$$\left|\sum_{r>R} \frac{z^r}{r} E(q^r)\right| \le \frac{|z|^{R+1}}{(R+1)(1-|z|)}. \tag{6.5}$$

The latter estimate uses $0 < E(q^r) < 1$ and a geometric series. Near $q = 1$, (6.2) is usually the more stable representation: its terms are positive after a change of sign when z is real and positive, while the original logarithmic sum exhibits severe cancellation.

6.3 Flatness at the endpoint

Put

$$t = -\log q > 0, \qquad A(x) = E(\mathrm{e}^{-x}) = \prod_{k\ge 0}(1 - \mathrm{e}^{-2^k x}), \tag{6.6}$$

and write

$$F(t,z) = -\log \Phi_{\mathrm{e}^{-t}}(z) = \sum_{r\ge 1} \frac{z^r}{r} A(rt). \tag{6.7}$$

For real $0 < z < 1$, this is a positive sum.

Theorem 6.2 (Uniform superalgebraic flatness)**.** *For every integer $M \ge 1$, every $0 < \rho < 1$, and all $t > 0$,*

$$\sup_{|z|\le\rho} |F(t,z)| \le 2^{M(M-1)/2} t^M \sum_{r\ge 1} r^{M-1}\rho^r. \tag{6.8}$$

In particular, on compact subsets of $|z| < 1$, $\Phi_{\mathrm{e}^{-t}}(z) - 1 = O(t^M)$ for every M. For fixed real $0 < z < 1$, extension by $\Phi_1(z) = 1$ cannot be real analytic in t at zero.

Proof. Keep only the first M factors of $A(x)$, bound every remaining factor by one, and use $1 - \mathrm{e}^{-u} \le u$:

$$0 < A(x) \le \prod_{k=0}^{M-1}(2^k x) = 2^{M(M-1)/2} x^M. \tag{6.9}$$

Substitute this in (6.7) and take absolute values. The resulting series in r converges because $\rho < 1$. Since $F \to 0$, exponentiation preserves the every-order estimate for $\Phi - 1 = \mathrm{e}^{-F} - 1$.

If the real extension were analytic, the every-order estimate would force all its positive-degree Taylor coefficients to vanish. It would then equal one in a neighborhood of zero. But Theorem 6.1 gives $\Phi_{\mathrm{e}^{-t}}(z) < 1$ for every $t > 0$. This contradiction proves nonanalyticity. □

The statement is stronger than any fixed power law, but it does not yet tell us how small F actually is. Optimizing (6.8) is possible, but a precise answer requires the logarithmic oscillation hidden inside A, together with a moving saddle over the index r.

7 An exact periodic amplitude and its Fabius connection

7.1 A complementary Laplace product

Let

$$h(x) = \frac{1 - \mathrm{e}^{-x}}{x}, \quad h(0) = 1, \qquad Q(x) = \prod_{k=1}^{\infty} h(2^{-k}x). \tag{7.1}$$

On compact sets, $h(2^{-k}x) = 1 + O(2^{-k})$, so the product converges locally uniformly and defines an entire function. For $x \geq 0$, all factors are positive.

Proposition 7.1 (A probability interpretation with fixed normalization)**.** *Let $U_1, U_2, \ldots$ be independent uniform random variables on $[0, 1]$ and set*

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k \in [0, 1]. \tag{7.2}$$

Then

$$Q(x) = \mathbb{E}\mathrm{e}^{-xX}. \tag{7.3}$$

If $\mathcal{F}(y) = \mathbb{P}(X \leq y)$, extended by zero for $y \leq 0$ and one for $y \geq 1$, then

$$\mathcal{F}'(y) = 2\big(\mathcal{F}(2y) - \mathcal{F}(2y - 1)\big). \tag{7.4}$$

In particular, $\mathcal{F}'(y) = 2\mathcal{F}(2y)$ for $0 < y < 1/2$.

Proof. The random series converges absolutely and is bounded by one. For a uniform variable U, $\mathbb{E}\mathrm{e}^{-sU} = h(s)$. Independence proves the Laplace identity for finite sums, and bounded convergence gives (7.3).

In distribution, $X = (U + X')/2$, where U is uniform on $[0, 1]$ and X' is an independent copy of X. Conditioning on X', or convolving with the uniform density, gives the density $2\mathbb{P}(2y - 1 \leq X' \leq 2y)$. The distribution has no atoms, because it contains an independent continuous uniform summand. Its distribution function is continuous, so this density is continuous and equals the derivative of $\mathcal{F}$. This proves (7.4). □

The dyadic uniform-convolution representation is the Fabius connection used here. Defining X and its distribution explicitly avoids the factor-of-two ambiguities that can arise between a Fabius distribution function, its density, and a centered Rvachev up-function. No unspoken normalization is needed.

7.2 The exact scaling invariant

Let $a = \log 2$. Directly from the products,

$$A(x/2) = (1 - \mathrm{e}^{-x/2})A(x), \qquad Q(x/2) = \frac{Q(x)}{h(x/2)}. \tag{7.5}$$

Thus $B(x) = A(x)Q(x)$ satisfies the simple equation

$$B(x/2) = \frac{x}{2} B(x). \tag{7.6}$$

Theorem 7.2 (Exact positive periodic factorization)**.** *Define, for real v,*

$$\Psi(v) = \exp\left(\frac{a}{2}(v^2 + v)\right) A(2^{-v})Q(2^{-v}). \tag{7.7}$$

Then Ψ is positive, smooth, and one-periodic. It and all its derivatives are bounded, and its minimum is positive. For every $x > 0$, writing $u = \log(1/x)$, one has the exact identity

$$A(x) = \exp\left(-\frac{u^2}{2a} - \frac{u}{2}\right) \frac{\Psi(u/a)}{Q(x)}. \tag{7.8}$$

Proof. The products defining A and all its derivatives converge uniformly on compact positive intervals; factors at large k have exponentially small errors in 2^k. The corresponding assertion for Q follows from its locally uniform entire product. Both functions are positive on the positive axis, so Ψ is positive and smooth.

Substitute $x = 2^{-v}$ in (7.6). The change from v to $v+1$ multiplies $B(2^{-v})$ by 2^{-v-1}, while the exponential factor in (7.7) is multiplied by $\exp(a(v+1))$. These factors cancel, proving periodicity. Boundedness of all derivatives and a positive minimum follow by restricting to the compact interval $[0,1]$. Finally, solve (7.7) for $A(x)$. □

The purpose of Q is not merely to approximate a product tail. Multiplying by it turns the dyadic functional equation into an exact scaling law with a linear factor. The remaining dependence on logarithmic scale is entirely periodic.

7.3 A convergent Bernoulli correction

Let the Bernoulli numbers be defined by $x/(\mathrm{e}^x - 1) = \sum_{n\ge 0} B_n x^n/n!$, so $B_1 = -1/2$.

Proposition 7.3 (Local expansion of the complementary factor)**.** *For $|x| < 2\pi$, using the logarithm that vanishes at zero,*

$$\log Q(x) = -\frac{x}{2} + \sum_{r\ge 1} \frac{B_{2r} x^{2r}}{2r(2r)!(2^{2r}-1)}. \tag{7.9}$$

In particular,

$$\log Q(x) = -\frac{x}{2} + \frac{x^2}{72} - \frac{x^4}{43200} + O(x^6). \tag{7.10}$$

Proof. The logarithmic derivative of h is

$$\frac{\mathrm{d}}{\mathrm{d}x} \log h(x) = \frac{1}{\mathrm{e}^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r\ge 1} \frac{B_{2r} x^{2r-1}}{(2r)!}.$$

Integrate from zero and substitute $x2^{-k}$. The linear terms sum using $\sum_{k\ge 1} 2^{-k} = 1$, and the even terms use $\sum_{k\ge 1} 2^{-2rk} = 1/(2^{2r}-1)$. Absolute local convergence permits the interchange of sums. Inserting $B_2 = 1/6$ and $B_4 = -1/30$ gives the first terms. □

The displayed radius is a convenient sufficient domain, not a claim of an optimal radius. Combining (7.8) and (7.9) gives an exact periodic factor times a convergent analytic correction near the endpoint.

7.4 Smooth flatness, not just a value estimate

Corollary 7.4 (The full zero jet)**.** *For every $|z| < 1$, the function $F(t,z)$, extended by $F(0,z) = 0$, is C^∞ for real $t \ge 0$ near zero. All its t-derivatives at zero vanish. The assertions and every-order bounds for each derivative are uniform for $|z| \le \rho < 1$. Consequently $\Phi_{\mathrm{e}^{-t}}(z)$, extended by one at zero, has the constant jet one there.*

Proof. Differentiate (7.8) a fixed number k of times. Since Ψ has bounded derivatives and $1/Q$ is analytic near zero, every resulting term is bounded, for small positive x, by

$$C_k x^{-k}(1 + |\log x|)^{d_k} \exp\left(-\frac{\log^2(1/x)}{2a} - \frac{\log(1/x)}{2}\right)$$

for some integer d_k. This is $O(x^M)$ for every M, because a negative quadratic in $\log(1/x)$ dominates every linear term and logarithm. On $x \ge \delta > 0$, product differentiation in (6.6) shows that $A^{(k)}(x)$ is bounded; the sums of derivative bounds are dominated by series of the form $\sum_j 2^{jk}\mathrm{e}^{-\delta 2^j}$. Enlarging the constant therefore gives, for every integer $M \geq 1$,

$$|A^{(k)}(x)| \leq C_{k,M}x^M \qquad (x > 0).$$

Termwise differentiation of (6.7) is valid away from zero, and

$$|\partial_t^k F(t,z)| \leq C_{k,M}t^M \sum_{r\geq 1} \rho^r r^{M+k-1}.$$

All these derivatives tend to zero. Extending them by zero and applying the fundamental theorem of calculus successively proves smoothness at the endpoint and the claimed derivatives. Exponentiation gives the final assertion. □

8 An all-order Lambert-W saddle expansion

8.1 Statement of the asymptotic law

We now fix $z \in (0,1)$ and put

$$\beta = -\log z > 0, \qquad L = \log(1/t), \qquad a = \log 2. \tag{8.1}$$

The principal Lambert function W_0 is the positive solution of $we^w = y$ when $y > 0$; this convention agrees with DLMF §4.13 [8]. Define

$$w = W_0\left(\frac{a\beta}{\sqrt{2}\,t}\right), \qquad r_* = \frac{w}{a\beta}, \qquad \theta = \frac{w}{a} + \frac{1}{2}. \tag{8.2}$$

All these parameters are real, and $w, r_* \to \infty$ as $t \to 0^+$. Set

$$P(t,z) = \frac{t\mathrm{e}^{a/8}}{\beta}\sqrt{\frac{2\pi w}{a}}\exp\left(-\frac{w^2+2w}{2a}\right). \tag{8.3}$$

For real θ, define the smooth amplitude on $y > 0$

$$\mathcal{A}_\theta(y) = \exp\left(-\frac{(\log y)^2}{2a}\right)\Psi\left(\theta - \frac{\log y}{a}\right). \tag{8.4}$$

Its derivative $\mathcal{A}_\theta^{(\ell)}(1)$ always means differentiation with respect to y; primes on Ψ mean differentiation with respect to its phase.

Theorem 8.1 (All-order saddle coefficients)**.** *For every fixed integer $K \geq 1$, as $t \to 0^+$,*

$$F(t,z) = P(t,z)\left[\sum_{n=0}^{K-1}\left(\frac{a}{w}\right)^n c_n(\theta) + O(w^{-K})\right]. \tag{8.5}$$

The estimate is uniform for z in a compact subset of $(0,1)$. Each c_n is a smooth one-periodic function. A finite formula for every coefficient is

$$c_n(\theta) = \sum_{\substack{\ell,k_3,\ldots,k_{2n+2}\geq 0\\ \ell+\sum_{j=3}^{2n+2}(j-2)k_j=2n}} \frac{\mathcal{A}_\theta^{(\ell)}(1)}{\ell!}\prod_{j=3}^{2n+2}\frac{1}{k_j!}\left(\frac{(-1)^{j+1}}{j}\right)^{k_j} \cdot\left(\ell + \sum_{j=3}^{2n+2} jk_j - 1\right)!!. \tag{8.6}$$

Here $(-1)!! = 1$*, and empty products have value one. In particular,*

$$c_0(\theta) = \Psi(\theta), \tag{8.7}$$

and

$$F(t,z) = P(t,z)\left[\Psi(\theta) + \frac{1}{w}\left\{\left(\frac{a}{12} - \frac{1}{2}\right)\Psi(\theta) - \frac{1}{2}\Psi'(\theta) + \frac{1}{2a}\Psi''(\theta)\right\} + O(w^{-2})\right]. \tag{8.8}$$

This is an all-order Poincaré expansion in $1/w$, with a bounded periodic coefficient field. It is not asserted to be a complete resurgent transseries: contributions smaller than every power of $1/w$ are absorbed into the remainder analysis, rather than individually enumerated.

8.2 Locating the saddle exactly

The proof begins by discarding only terms that are beyond every algebraic order in $1/w$. For $r \le t^{-1/2}$, (7.9) gives $Q(rt)^{-1} = 1 + O(\sqrt{t})$, uniformly. For $r > t^{-1/2}$, $A(rt) \le 1$ and the geometric tail in (6.5) is bounded by $O(\exp(-\beta/\sqrt{t}))$, uniformly when β is in a compact positive interval. The eventual saddle main term has logarithm of order $-L^2$, so this tail is negligible to every relative power of $1/w$. A lower bound of that size can already be obtained from the summand nearest r_*, using the positive minimum of Ψ.

Using (7.8), the main part of the summand is

$$\exp(f(r))\Psi\left(\frac{L - \log r}{a}\right), \qquad f(r) = -\beta r - \frac{(L - \log r)^2}{2a} - \frac{L - \log r}{2} - \log r. \tag{8.9}$$

Its exponential phase has derivative

$$f'(r) = -\beta + \frac{L - \log r}{ar} - \frac{1}{2r}.$$

Thus $f'(r) = 0$ is equivalent to

$$a\beta r + \log r = L - \frac{a}{2}. \tag{8.10}$$

The left side is strictly increasing on $(0, \infty)$, so there is exactly one solution. With $w = a\beta r$, exponentiation turns (8.10) into $we^w = a\beta/(\sqrt{2}t)$, giving (8.2). Notice that we locate the saddle of the large exponential factor; the periodic amplitude is retained in the coefficient calculus and need not be absorbed into a shifted saddle.

At the saddle,

$$L - \log r_* = w + \frac{a}{2}, \qquad f(r_*) = -\frac{w^2 + 2w}{2a} + \frac{a}{8} - L. \tag{8.11}$$

An especially useful identity holds for every $y > 0$:

$$f(r_* y) - f(r_*) = \frac{w}{a}(\log y - y + 1) - \frac{(\log y)^2}{2a}. \tag{8.12}$$

Unlike a local Taylor expansion, this is an exact global identity. The function $H(y) = \log y - y + 1$ is nonpositive, has its unique maximum zero at $y = 1$, and satisfies $H''(1) = -1$. It therefore gives both localization and the Gaussian scale.

8.3 Why the lattice does not change the algebraic coefficients

For clarity, the sum-to-integral step is justified explicitly. Let χ be a smooth function of y supported in $(1/3, 3)$, equal to one on $[1/2, 2]$, and define

$$g(r) = \chi(r/r_*)\mathrm{e}^{f(r)}\Psi\left(\frac{L - \log r}{a}\right).$$

Outside the central interval, (8.12) bounds both the sum and integral tails by $\mathrm{e}^{f(r_*)}$ times a polynomial in w times e^{-cw}, for some positive c. For large y, the term $-wy/a$ provides a geometric or exponential tail; near zero, $y^{w/a}$ provides the corresponding integrability. The periodic amplitude is bounded. These estimates are uniform for β in a compact positive interval.

For a smooth compactly supported function on the real line and any integer $p \geq 2$, the Euler summation identity gives

$$\left|\sum_{n\in\mathbb{Z}} g(n) - \int_{\mathbb{R}} g(r)\,\mathrm{d}r\right| \leq C_p \int_{\mathbb{R}} |g^{(p)}(r)|\,\mathrm{d}r. \tag{8.13}$$

One way to see this bound is to integrate by parts p times against the periodic Bernoulli function in the Euler summation formula. Its boundedness produces the constant C_p; compact support removes all endpoint terms.

The peak width in the r variable is proportional to $\sqrt{w}$, since $r_* \asymp w$ and the y-width is $w^{-1/2}$. Differentiation of (8.12) on the central interval, followed by Gaussian domination, yields

$$\int_{\mathbb{R}} |g^{(p)}(r)|\,\mathrm{d}r \leq C'_p \mathrm{e}^{f(r_*)} w^{(1-p)/2}. \tag{8.14}$$

For completeness, each derivative of the exponential is a sum of products of phase derivatives. In the local variable $v = (r - r_*)/\sqrt{w}$, a derivative with respect to r contributes $w^{-1/2}$, while the resulting factors are bounded by polynomials in $|v|$ times e^{-cv^2}. Integrating $\mathrm{d}r = \sqrt{w}\,\mathrm{d}v$ gives (8.14). Away from a fixed central neighborhood, all cutoff derivatives are exponentially smaller and satisfy the same bound after enlarging the constant. Derivatives of Ψ are bounded and introduce factors $1/r = O(w^{-1})$, no larger than required.

The integral itself is comparable to $\mathrm{e}^{f(r_*)}\sqrt{w}$, using positivity of Ψ. Combining (8.13) and (8.14), the relative lattice error is $O(w^{-p/2})$. Since p can be chosen arbitrarily large, the lattice does not affect any fixed algebraic coefficient of the saddle expansion. This is an all-orders statement, not an assertion that the lattice error is exactly zero.

8.4 The Gaussian coefficient calculation

After the preceding reductions and the substitution $r = r_* y$, it remains to expand

$$\mathrm{e}^{f(r_*)} r_* \int_0^\infty \exp\left(k(\log y - y + 1)\right)\mathcal{A}_\theta(y)\,\mathrm{d}y, \qquad k = \frac{w}{a}. \tag{8.15}$$

Put $y = 1 + v/\sqrt{k}$. Then

$$k(\log y - y + 1) = -\frac{v^2}{2} + \sum_{j\geq 3} \frac{(-1)^{j+1} v^j}{j\, k^{(j-2)/2}}. \tag{8.16}$$

Expand the analytic phase correction and the smooth amplitude to the required finite order. A term from amplitude derivative ℓ and phase multiplicities $k_3, k_4, \ldots$ contributes the power

$$k^{-\frac{1}{2}\{\ell + \sum_{j\geq 3}(j-2)k_j\}}$$

and the monomial $v^{\ell+\sum jk_j}$. Terms of odd degree integrate to zero against the centered Gaussian. At power k^{-n}, the degree constraint is precisely the one in (8.6); the degree of the Gaussian monomial is even because it differs from $2n$ by $2\sum k_j$. Its normalized integral is the double factorial in that formula.

Here is a remainder justification at arbitrary fixed order. First restrict $|y-1| \le \delta$ with small fixed δ, where $H(y) \le -c(y-1)^2$. Outside this interval the integral is exponentially smaller. Inside it, restrict further to $|v| \le k^\epsilon$, where $\epsilon > 0$ is chosen small enough for the desired finite Taylor order. Taylor's theorem with a bounded derivative remainder applies uniformly in the periodic parameter θ. Expanding sufficiently many phase and amplitude terms and integrating polynomial remainders against e^{-cv^2} gives an error $O(k^{-K})$ after retaining powers through $k^{-(K-1)}$. The region $k^\epsilon < |v| \le \delta\sqrt{k}$ has a Gaussian tail smaller than every power of k^{-1}. The phase and amplitude bounds are uniform in θ, so the expansion is uniform. This proves the all-order coefficient formula without treating a divergent Taylor expansion as a global identity.

The Gaussian normalization in (8.15) is

$$\mathrm{e}^{f(r_*)} r_* \sqrt{\frac{2\pi a}{w}} = P(t,z),$$

by (8.2) and (8.11). This proves (8.5). Periodicity and smoothness of c_n follow from those of Ψ and the finite formula (8.6).

8.5 First and second universal coefficients

Writing $A_j = \mathcal{A}_\theta^{(j)}(1)$ temporarily, the coefficient formula gives

$$c_1 = \frac{A_0}{12} + A_1 + \frac{A_2}{2}, \tag{8.17}$$

$$c_2 = \frac{A_0}{288} + \frac{A_1}{12} + \frac{25A_2}{24} + \frac{5A_3}{6} + \frac{A_4}{8}. \tag{8.18}$$

These identities can also be checked directly by Gaussian moments of degrees up to twelve. They are included in the symbolic verification program.

The chain rule gives

$$A_0 = \Psi, \qquad A_1 = -\frac{\Psi'}{a}, \qquad A_2 = -\frac{\Psi}{a} + \frac{\Psi'}{a} + \frac{\Psi''}{a^2}, \tag{8.19}$$

where the functions on the right are evaluated at θ. Substitution in (8.17) gives (8.8). This completes the proof of Theorem 8.1.

8.6 Why the first term of the logarithmic series is misleading

A tempting approximation to (6.7) is to keep only $zA(t)$. It has the right qualitative flatness, but not the correct magnitude.

Corollary 8.2 (Collective enhancement of the flat tail)**.** *For fixed $z \in (0,1)$, as $L = \log(1/t) \to \infty$,*

$$\log F(t,z) = -\frac{L^2}{2a} + \frac{L\log L}{a} + O(L), \tag{8.20}$$

$$\log \frac{F(t,z)}{A(t)} = \frac{L\log L}{a} + O(L). \tag{8.21}$$

The contributing indices are centered at $r_* = w/(a\beta) \sim L/(a\beta)$, with width of order $\sqrt{L}$.

Proof. Taking logarithms in $we^w = a\beta/(\sqrt{2}t)$ gives $w + \log w = L + \log(a\beta/\sqrt{2})$. It follows first that $w \sim L$, and then, by substitution, that $w = L - \log L + O(1)$. The leading term of Theorem 8.1, together with the positive upper and lower bounds for Ψ, yields

$$\log F = -L - \frac{w^2 + 2w}{2a} + \frac{1}{2}\log w + O(1).$$

Substituting the estimate for w proves (8.20). Formula (7.8) gives $\log A(t) = -L^2/(2a) - L/2 + O(1)$, because $Q(t) = 1 + O(t)$. Subtracting proves (8.21). The location and width were derived in (8.2) and the lattice analysis. □

Every individual term is flat, but the effective index drifts to infinity. This is precisely why pointwise asymptotics at a fixed index cannot be interchanged with the infinite sum without a uniform argument.

8.7 Parameter regimes deliberately not covered

The expansion is uniform when z stays in a compact subset of $(0, 1)$. It does not assert uniformity as $z \to 0$, where the saddle may cease to be large under joint scaling, or as $z \to 1$, where $\beta \to 0$ and the geometric cutoff changes scale. For complex z, the defining product and flatness theorem remain valid, but positivity and the real saddle proof no longer apply as stated. Those cases require separate contour and cancellation analysis. In particular, the present proof does not license replacing z by a root of unity in (8.5).

9 A joint endpoint limit and classical Thue–Morse products

The fixed-z theorem has a particularly instructive companion. Approach the corner $(q, z) = (1, 1)$ along $z = q^\gamma$, with $\gamma > 0$ fixed. Here the product no longer tends to one. Instead, a classical sign-weighted product emerges, and every algebraic discretization correction vanishes.

Theorem 9.1 (A flat boundary layer). *For $\gamma > 0$, define*

$$\mathcal{H}(\gamma) = \int_0^\infty e^{-\gamma x} \frac{A(x)}{x}\, dx, \qquad \mathcal{P}(\gamma) = \exp(-\mathcal{H}(\gamma)). \tag{9.1}$$

The integral is finite and positive. For every integer $M \geq 1$,

$$F(t, e^{-\gamma t}) = \mathcal{H}(\gamma) + O(t^M), \tag{9.2}$$

$$\Phi_{e^{-t}}(e^{-\gamma t}) = \mathcal{P}(\gamma) + O(t^M) \tag{9.3}$$

as $t \to 0^+$, uniformly for γ in compact subsets of $(0, \infty)$. Moreover,

$$\mathcal{P}(\gamma) = \prod_{j=0}^{\infty} \left(\frac{\gamma + 2j}{\gamma + 2j + 1} \right)^{\varepsilon_j}. \tag{9.4}$$

The product on the right is understood in the displayed paired order.

Proof. The function $g_\gamma(x) = e^{-\gamma x} A(x)/x$, for $x > 0$, extends by zero to a smooth function on the whole real line. Smooth flatness of A, proved in Section 7, handles the origin and the factor $1/x$. At positive infinity, the factor $e^{-\gamma x}$ ensures integrability of every derivative. All derivative L^1 bounds are uniform for γ in a compact positive interval. In particular, g_γ is integrable, and its integral is strictly positive.

Formula (6.7) becomes the Riemann sum

$$F(t, e^{-\gamma t}) = t \sum_{r \geq 1} g_\gamma(rt). \tag{9.5}$$

The mesh-t version of (8.13) gives, for any integer $p \geq 2$,

$$\left| t \sum_{r\in\mathbb{Z}} g_\gamma(rt) - \int_{\mathbb{R}} g_\gamma(x)\,\mathrm{d}x \right| \leq C_p t^p \|g_\gamma^{(p)}\|_1.$$

One may first apply it to compactly supported cutoffs and then remove the cutoffs using exponential decay. The terms for nonpositive r are zero. Choosing $p \geq M$ proves (9.2), and exponentiation proves (9.3). Thus no nonzero Euler–Maclaurin endpoint coefficient survives: every derivative at the left endpoint is zero, and those at infinity vanish as well.

To identify the limiting product, use the finite dyadic polynomial

$$A_m(x) = \prod_{k=0}^{m-1} (1 - \mathrm{e}^{-2^k x}) = \sum_{n=0}^{2^m-1} \varepsilon_n \mathrm{e}^{-nx}.$$

For $m \geq 1$, $0 \leq A_m(x) \leq \min(x, 1)$. These products decrease to $A(x)$, and the displayed bound gives an integrable majorant for $\mathrm{e}^{-\gamma x} A_m(x)/x$. Since $\sum_{n<2^m} \varepsilon_n = 0$, Frullani's identity gives

$$\int_0^\infty \mathrm{e}^{-\gamma x} \frac{A_m(x)}{x}\,\mathrm{d}x = -\sum_{n<2^m} \varepsilon_n \log(\gamma + n). \tag{9.6}$$

For justification, replace each exponential by its difference from $\mathrm{e}^{-\gamma x}$, use the zero sum of signs, and integrate $\int_0^\infty (\mathrm{e}^{-ux} - \mathrm{e}^{-vx})\,\mathrm{d}x/x = \log(v/u)$. The latter identity follows by integrating $\mathrm{e}^{-ux} - \mathrm{e}^{-vx} = \int_u^v x\mathrm{e}^{-sx}\,\mathrm{d}s$ and interchanging a finite parameter integral with the positive exponential integral.

Dominated convergence in (9.6), followed by exponentiation, shows that $\mathcal{P}(\gamma)$ is the limit of $\prod_{n<2^m} (\gamma + n)^{\varepsilon_n}$. Pairing indices $2j$ and $2j + 1$ converts it into partial products of (9.4). Their logarithmic series converges: the partial sums of ε_j are bounded, since consecutive even–odd signs cancel, while $\log((\gamma + 2j)/(\gamma + 2j + 1))$ tends monotonically to zero. Dirichlet summation therefore proves convergence in the paired order and identifies its limit with the dyadic subsequence. □

Remark 9.2 (The ordering is essential). The unpaired formal series $\sum_{n\geq 0} \varepsilon_n \log(\gamma + n)$ does not converge in its ordinary term-by-term order: its terms do not tend to zero. Formula (9.4) is a convergent *paired* product, and (9.6) uses balanced dyadic blocks. Omitting this distinction would turn a correct limiting identity into a false convergence claim.

9.1 Two exact constants

The boundary-layer integral permits a short proof of two special values.

Proposition 9.3. *For* $\gamma > 0$,

$$\mathcal{H}(\gamma/2) - \mathcal{H}((\gamma + 1)/2) = \mathcal{H}(\gamma). \tag{9.7}$$

Furthermore,

$$\mathcal{H}(1/2) = \log 2, \qquad \mathcal{H}(1) = \frac{1}{2}\log 2, \qquad \mathcal{P}(1/2) = \frac{1}{2}, \quad \mathcal{P}(1) = \frac{1}{\sqrt{2}}. \tag{9.8}$$

Consequently, for every fixed M*,*

$$\Phi_q(q) = \frac{1}{\sqrt{2}} + O((-\log q)^M), \qquad \Phi_q(\sqrt{q}) = \frac{1}{2} + O((-\log q)^M) \quad (q \to 1^-). \tag{9.9}$$

Proof. Subtract the two integrals on the left of (9.7). The numerator contains $\mathrm{e}^{-\gamma x/2}(1 - \mathrm{e}^{-x/2})A(x) = \mathrm{e}^{-\gamma x/2}A(x/2)$, by (7.5). The substitution $x = 2y$ gives $\mathcal{H}(\gamma)$.

Also, $A(x) = (1 - \mathrm{e}^{-x})A(2x)$, so

$$A(2x) - A(x) = \mathrm{e}^{-x}A(2x).$$

Integrate its quotient by x. On $[\delta, R]$, the left side integrates to

$$\int_R^{2R} \frac{A(y)}{y}\,\mathrm{d}y - \int_\delta^{2\delta} \frac{A(y)}{y}\,\mathrm{d}y.$$

As $R \to \infty$ and $\delta \to 0$, this tends to $\log 2$, because $A(y) \to 1$ at infinity and $A(y) \to 0$ at zero. The right side, under $y = 2x$, integrates to $\mathcal{H}(1/2)$. Thus $\mathcal{H}(1/2) = \log 2$. Taking $\gamma = 1$ in (9.7) gives $\mathcal{H}(1/2) = 2\mathcal{H}(1)$. The remaining formulas follow by exponentiation and Theorem 9.1. □

The paired product at $\gamma = 1$ is the classical Woods–Robbins Thue–Morse product, not a newly discovered constant [5, 2]. The derived refinement here is its realization as a joint endpoint value of $\Phi_q(z)$, with an error smaller than every power of $-\log q$. It also explains concretely why the fixed-z limit cannot be uniform up to $z = 1$.

9.2 Monotonicity of the boundary profile

Differentiation under the integral gives, for every $k \geq 1$,

$$(-1)^k\mathcal{H}^{(k)}(\gamma) = \int_0^\infty x^{k-1}\mathrm{e}^{-\gamma x}A(x)\,\mathrm{d}x > 0. \tag{9.10}$$

Thus $\mathcal{H}$ is strictly decreasing and convex, and $\mathcal{P}$ is strictly increasing. Dominated convergence gives $\mathcal{H}(\gamma) \to 0$ as $\gamma \to \infty$. Since $A(x) \to 1$, integrating over a large fixed lower endpoint up to $1/\gamma$ shows $\mathcal{H}(\gamma) \to \infty$ as $\gamma \to 0^+$. The profile $\mathcal{P}$ therefore increases from zero to one. This is a continuous family of limiting constants, rather than an isolated diagonal coincidence.

10 Reproducible symbolic and numerical checks

10.1 What was checked exactly

The supplied program `verify.py` uses integers and rational arithmetic for the finite combinatorial identities. The deterministic random seed is 20260904 for the primary tests; the additional block tests use 410926. It performs 1,800 assertions on even-order dyadic correlation identities, using 120 tuples of lengths $2, 4, 6, 8$ and several consecutive levels. A further 960 assertions test odd-order sums, cyclic cancellation, and the block-multiple formulas in (3.14).

The covering identity is compared coefficient by coefficient with direct enumeration of the Boolean cube, giving 35 exact moment equalities in the selected small examples. Nine path examples test (4.14)–(4.15), and six symbolic tests compare the weak-coupling coefficient with the independently enumerated matching polynomial. The general coefficient generator reproduces (8.17)–(8.18).

All these assertions passed. They are finite checks of the implementation and formulas, not replacements for the general proofs.

10.2 Saddle verification at very small scales

Numerical computations use 80 decimal digits of working precision. We evaluate $F = -\log\Phi$, rather than subtracting Φ from one: in the last row of Table 3, the deviation has size about 10^{-1824}, far below a direct 80-digit subtraction from one.

Table 2: The first nonzero signed moment of the path coordinate (4.13) at $\lambda = 1$.

Number of bits m	First degree	First nonzero moment
2	1	1
3	2	−12
4	2	2
5	3	−138
6	3	6
7	4	−2112
8	4	24
9	5	−41400
10	5	120

The product $A(x)$ is evaluated through the stable logarithmic expression $\log(1-\mathrm{e}^{-y}) = \log(-\operatorname{expm1}(-y))$, doubling y at successive factors. The positive series (6.7) is truncated with the explicit bound (6.5). The periodic amplitude is evaluated on one period using (7.7) and the convergent series (7.9).

Table 3: Saddle approximations for $z = 1/2$, $t = \mathrm{e}^{-L}$. A signed relative error means approximation divided by the positive series value, minus one.

L	w	$\log_{10} F$	Leading error	With first correction
10	6.977684	−23.376724	$5.8521 \cdot 10^{-2}$	$-8.5644 \cdot 10^{-3}$
20	16.139152	−99.611573	$2.6520 \cdot 10^{-2}$	$-1.6113 \cdot 10^{-3}$
40	35.354962	−430.149058	$1.2320 \cdot 10^{-2}$	$-3.4163 \cdot 10^{-4}$
80	74.608151	−1824.180963	$5.8848 \cdot 10^{-3}$	$-7.7430 \cdot 10^{-5}$

The omitted positive-series tails, divided by the computed values, have bounds below $1.5 \cdot 10^{-75}$ in these examples. This bounds only the omitted r-terms, not every floating-point or product-evaluation error. The computation is not an interval-certified enclosure. Independent checks of the defining logarithmic product against the positive series at moderate parameters agree to at least 65 decimal places. The exact periodic factorization likewise agrees at the working precision in the tested cases; a printed residual of zero means numerical agreement, not an additional exact proof.

10.3 The diagonal boundary layer

For $z = q = \mathrm{e}^{-t}$, direct logarithmic-product evaluation gives Table 4. The signs of the errors need not be monotone; the statement is a superalgebraic bound, not monotone convergence of the error.

Table 4: The joint endpoint $\Phi_q(q) \to 1/\sqrt{2}$. Values are rounded; the full data are in `boundary_layer.csv`.

t	$\Phi_{\mathrm{e}^{-t}}(\mathrm{e}^{-t})$	Error from $1/\sqrt{2}$
1	0.758123053523105	$5.1016 \cdot 10^{-2}$
0.5	0.710249881663610	$3.1431 \cdot 10^{-3}$
0.25	0.706267615662706	$-8.3917 \cdot 10^{-4}$
0.125	0.707016842592727	$-8.9939 \cdot 10^{-5}$
0.0625	0.707106345877474	$-4.3531 \cdot 10^{-7}$

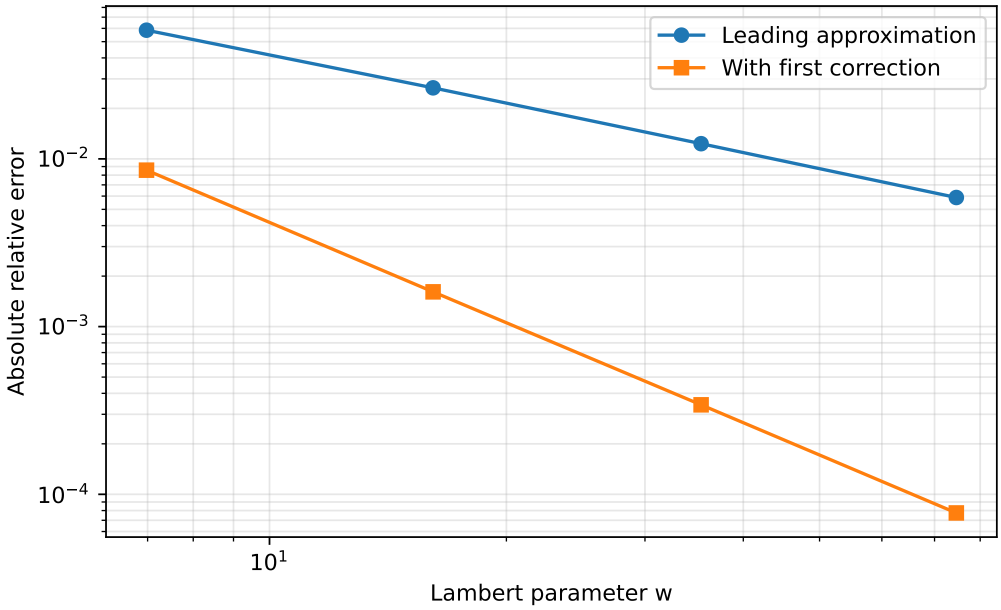

Figure 1: Absolute relative errors for the data in Table 3. The first correction improves the observed error from the scale $1/w$ to the scale $1/w^2$, as predicted by the theorem. The graph is a numerical check, not a uniform error bound.

10.4 How to reproduce the archive

The source is self-contained apart from the included vector figure. Run `latexmk -pdf thue_morse_new_directions.tex`, or run `pdflatex` three times. The verification program requires Python 3.10 or later, `mpmath`, and `sympy`. Adding `-plots` also uses `matplotlib` and regenerates the figure. Run `python verify.py -plots` from the article directory. The program writes its JSON summary and CSV tables beside itself.

The exact checks and floating-point checks are separated in the output. A failed assertion stops the computation rather than silently producing a success report. The numerical routines are intended for the explicitly stated real parameter ranges; they are not implementations of analytic continuation or a general complex saddle solver.

11 Further deductions, limitations, and research directions

11.1 A unifying finite–infinite correspondence

There are three distinct forms of compression in the proved results. In the correlation problem, all interactions with a dyadic boundary are compressed into the integer $D(\boldsymbol{h})$. In the nonlinear moment problem, all Boolean cancellations are compressed into the covering condition $\bigcup \mathcal{C} = V$. In the product problem, the exact dyadic scaling is compressed into the periodic amplitude Ψ, while the continuous part is handled by one saddle.

The comparison is structural, not an assertion that the three formulas are special cases of a single theorem already proved here. Their common feature is that cancellation is dealt with *before* taking the asymptotic limit. Doing the reverse can lose the boundary term, misidentify the first surviving moment, or retain the wrong index in an infinite sum.

11.2 Classifying vanishing boundary functionals

Research question 11.1. Which tuples $\boldsymbol{h}$, modulo translations and deletion of repeated pairs, satisfy $D(\boldsymbol{h}) = 0$? Can their binary structure be characterized without explicitly summing (3.5)?

For even order, $D = 0$ means every sufficiently large dyadic cyclic and noncyclic sum equals its mean times the block length exactly. For odd order, it means every sufficiently large aligned dyadic block sum is zero. This is a finite classification problem with a precise observable consequence. The pair recurrence (3.20) provides a starting model, but the higher-order combinatorics of the threshold parities b_t are richer. No classification is claimed in this article.

11.3 Designed nonlinear cancellation beyond a hafnian

Research question 11.2. For a prescribed sparse interaction graph, what is the largest possible vanishing order of $Z_x(t)$ among nontrivial signed quadratic weights, subject to chosen normalization constraints on the linear part?

The positive-weight answer is determined by the minimum cover. Signed weights allow algebraic cancellation among minimum covers, so the problem becomes a finite polynomial system built from (4.6). Useful constraints would exclude degenerate choices that make x independent of a bit and force $Z_x \equiv 0$. Requiring the coordinate map to be injective on the Boolean cube is another natural way to rule out trivial pairwise coincidences. The article supplies the equations but does not assert an optimal solution.

11.4 Uniform matching of the two analytic regimes

The fixed-z saddle corresponds to $\beta = -\log z$ bounded away from zero, while the boundary layer has $\beta = \gamma t$. The intermediate regime $t \ll \beta \ll 1$ should connect the large-γ asymptotics of $\mathcal{H}(\gamma)$ with the moving discrete saddle. A rigorous uniform formula would need to control the relation between saddle width and mesh size throughout that transition. Theorems 8.1 and 9.1 establish the two endpoints but do not prove the full uniform bridge.

Research question 11.3. Can one construct a single asymptotic approximation, with an explicit uniform remainder, valid when $\beta/t \to \infty$ while $\beta \to 0$, and matching both (8.5) and (9.2)?

This is a concrete continuation problem rather than a request to speculate about a nonexistent singularity. The exact positive representation (6.7) and the amplitude identity (7.8) remain valid throughout the real parameter region and provide the required starting point.

11.5 Beyond-all-orders corrections and complex parameters

The saddle theorem suppresses lattice effects beyond every algebraic order in $1/w$. The boundary-layer theorem suppresses all Euler summation powers of t. Identifying the leading remaining corrections would require more than extending the coefficient list: one must study the analytic continuation of the amplitude and the relevant complex saddles or Fourier modes. That problem is a natural transseries extension, but it is not solved by the present all-order Poincaré expansion. Similarly, $|z| = 1$ is a boundary of the absolute convergence arguments, and root-of-unity specialization requires a fresh proof.

11.6 Possible formalization targets

The finite correlation and covering theorems have a relatively direct route to formal proof: finite sums, bit decompositions, parity, and Boolean monomial identities suffice. An initial formalization could isolate (3.10), (3.11), and (4.3), then derive the remaining finite identities as corollaries. Such a project should use the actual theorem names and conventions of the current repository after a separate code audit; none are invented here.

The analytic results require additional infrastructure for infinite products, termwise differentiation, dominated convergence, and uniform saddle estimates. They should not be marked formalized merely because their numerical checks pass. The supplied Python code is reproducibility material, not a proof assistant certificate.

12 Conclusion

Three extensions of the Thue–Morse calculus have been obtained with explicit formulas and proofs. The correlation theory admits a finite boundary functional that gives exact defects at every order, including an odd-order telescoping law and endpoint formulas involving an alternating binary digit sum. Nonlinear Prouhet sums admit an exact hypergraph-cover expansion, making graph matchings and hafnians the first surviving moment data of quadratic deformations. The automatic geometric product $\Phi_q(z)$ exhibits smooth flatness, a periodic Lambert-W saddle law to every algebraic order, and a joint endpoint profile that recovers classical paired Thue–Morse products with superalgebraically small errors.

The results are concrete enough to test, extend, and formalize. Their historical novelty remains a separate question from their mathematical validity: the proofs establish the formulas, while the source audit provides a bounded, explicitly stated comparison with existing work.

A Coefficient generation without hidden recurrences

For a requested saddle order n, formula (8.6) is a finite weighted partition sum. An implementation can loop over k_j with $0 \le k_j \le \lfloor 2n/(j-2) \rfloor$, discard assignments for which $\sum (j-2)k_j > 2n$, and set $\ell = 2n - \sum (j-2)k_j$. This directly generates the coefficient in the formal symbols $A_0, \dots, A_{2n}$. The program's recursive enumeration is only a traversal of this finite index set; the mathematical coefficient formula does not depend on previously computed coefficients.

The derivatives A_j can themselves be expressed as finite combinations of $\Psi^{(r)}(\theta)$. Let $s = \log y$ and

$$B_\theta(s) = \mathrm{e}^{-s^2/(2a)} \Psi(\theta - s/a).$$

Since $\mathrm{d}/\mathrm{d}y = \mathrm{e}^{-s}\, \mathrm{d}/\mathrm{d}s$, induction gives

$$\mathcal{A}_\theta^{(\ell)}(1) = \left(\prod_{j=0}^{\ell-1} (D_s - j) \right) B_\theta(s) \bigg|_{s=0}, \qquad D_s = \frac{\mathrm{d}}{\mathrm{d}s}. \tag{A.1}$$

For $\ell = 0$, the empty operator product is the identity. To prove the formula, write $\mathrm{d}^\ell/\mathrm{d}y^\ell = \mathrm{e}^{-\ell s} \prod_{j=0}^{\ell-1} (D_s - j)$ and differentiate once more; the derivative of $\mathrm{e}^{-\ell s}$ introduces the next factor $D_s - \ell$.

If $s(\ell, j)$ denotes the signed Stirling coefficient defined by $u(u-1)\cdots(u-\ell+1) = \sum_j s(\ell, j) u^j$, then

$$\mathcal{A}_\theta^{(\ell)}(1) = \sum_{j=0}^{\ell} s(\ell, j) \sum_{h=0}^{\lfloor j/2 \rfloor} \frac{j!}{h!(j-2h)!} \left(-\frac{1}{2a} \right)^h \left(-\frac{1}{a} \right)^{j-2h} \Psi^{(j-2h)}(\theta). \tag{A.2}$$

Indeed, expand the Gaussian factor in even powers of s, expand the translated Ψ by its finite derivative coefficient, and extract the j-th derivative at zero. Substituting (A.2) into (8.6) produces a fully explicit finite formula for each saddle coefficient in the derivatives of the periodic function.

B Notation and normalization reference

Symbol	Meaning
ε_n	The sign $(-1)^{s_2(n)}$, with $\varepsilon_0 = 1$.
$E(z)$	The ordinary generating function $\sum_{n\geq 0} \varepsilon_n z^n$.
$\boldsymbol{h}, H, p$	A shift tuple with minimum zero, its maximum, and its length.
$D(\boldsymbol{h})$	The integer boundary functional in (3.5).
S_m, R_m	Noncyclic and cyclic sums over a block of length 2^m.
$\sigma(Q)$	The alternating binary digit sum, not the ordinary digit sum.
$Z_x(t)$	The signed exponential sum on a nonlinearly positioned Boolean cube.
$\mathcal{H}, \tau$	The family of nonempty monomial supports and its minimum cover size.
$\mathrm{Haf}(J)$	The sum of products over perfect matchings; no determinant signs.
$\mathcal{M}_G$	The weighted matching polynomial, including unmatched singleton weights.
$\Phi_q(z)$	The geometric product $\prod_{j\geq 0}(1 - zq^j)^{\varepsilon_j}$.
$A(x)$	The positive product $E(\mathrm{e}^{-x})$.
$Q(x)$	The Laplace transform of $\sum_{k\geq 1} 2^{-k} U_k$, with U_k uniform on $[0, 1]$.
$\Psi(v)$	The positive one-periodic function in (7.7).
$F(t, z)$	The positive logarithmic deviation $-\log \Phi_{\mathrm{e}^{-t}}(z)$ for real positive z.
a, β, L	Respectively $\log 2$, $-\log z$, and $\log(1/t)$.
w, r_*, θ	The Lambert parameter, saddle location, and periodic phase in (8.2).
$P(t, z), c_n$	The saddle prefactor and its periodic coefficient functions.
$\mathcal{H}, \mathcal{P}$	The joint-endpoint integral and its exponential, distinct from the saddle prefactor $P(t, z)$.

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